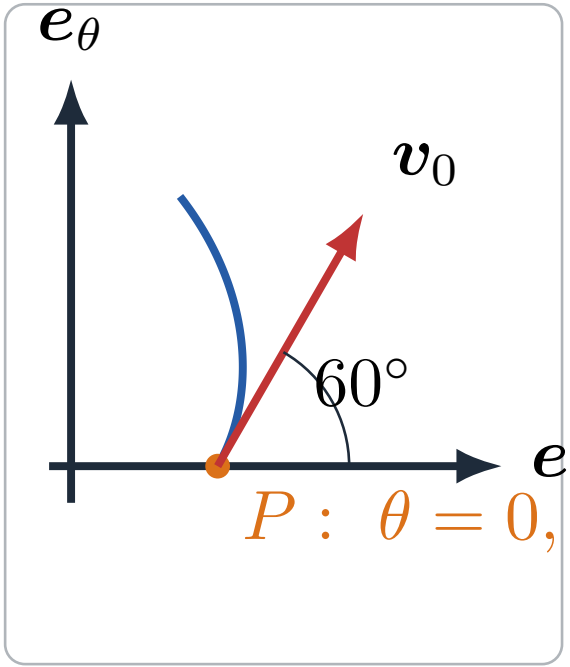
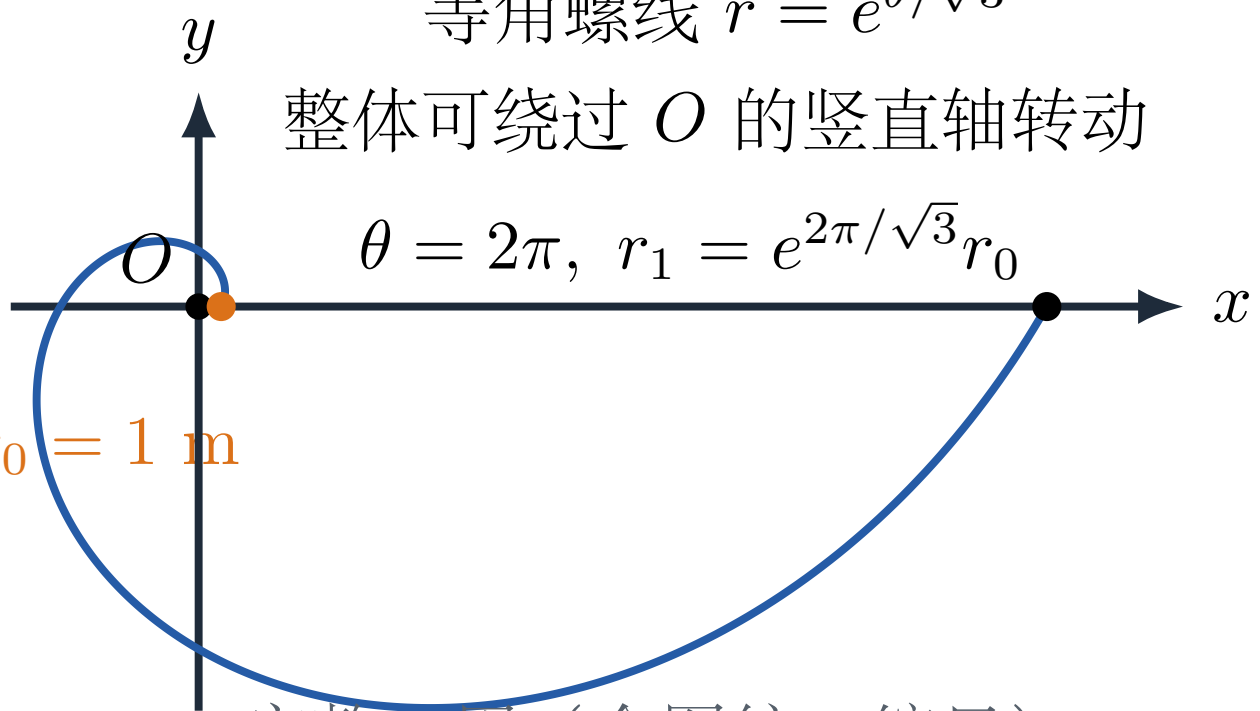


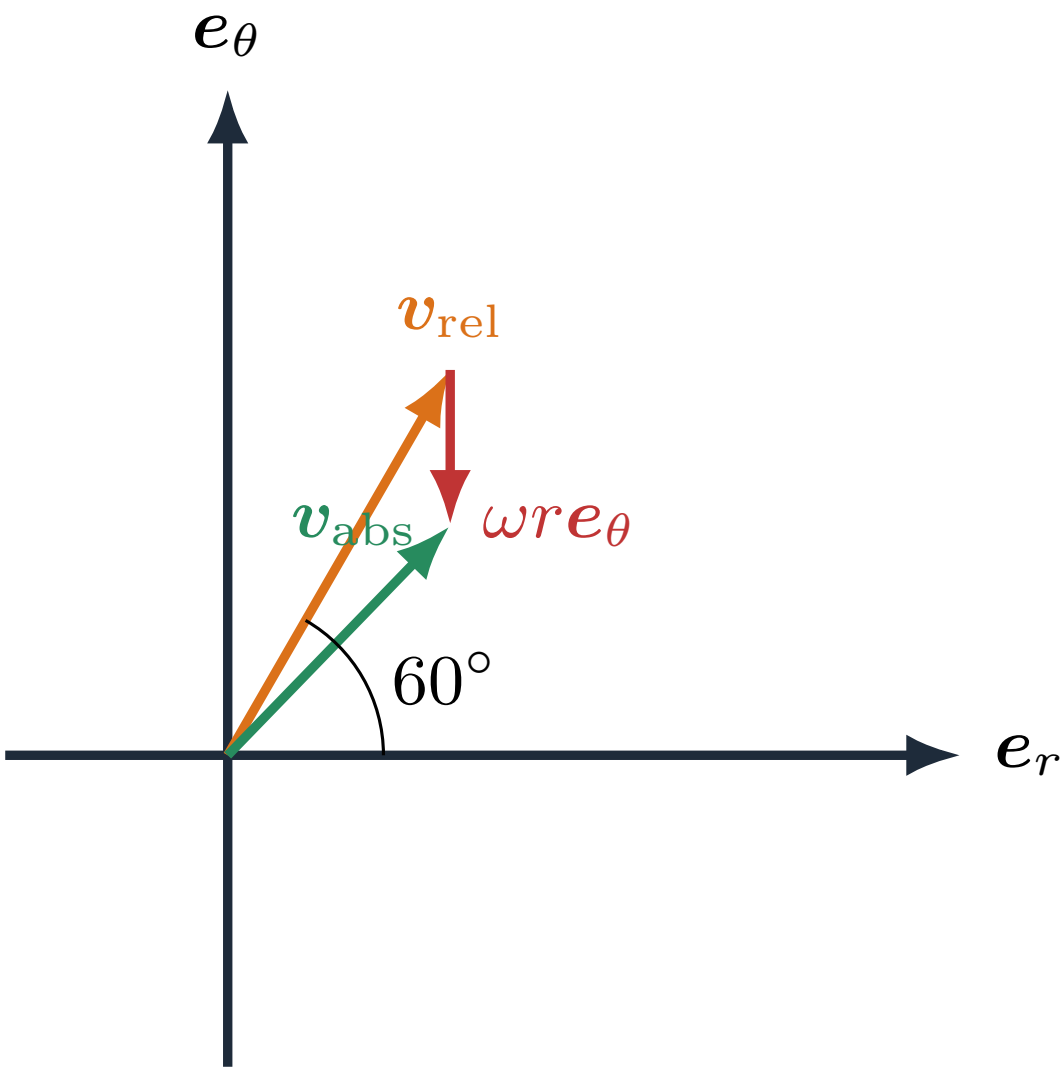
等角螺线 $r = e^{\theta/\sqrt{3}}$

整体可绕过 O 的竖直轴转动

$\theta = 2\pi,\ r_1 = e^{2\pi/\sqrt{3}} r_0$



完整一周（全图统一缩尺）

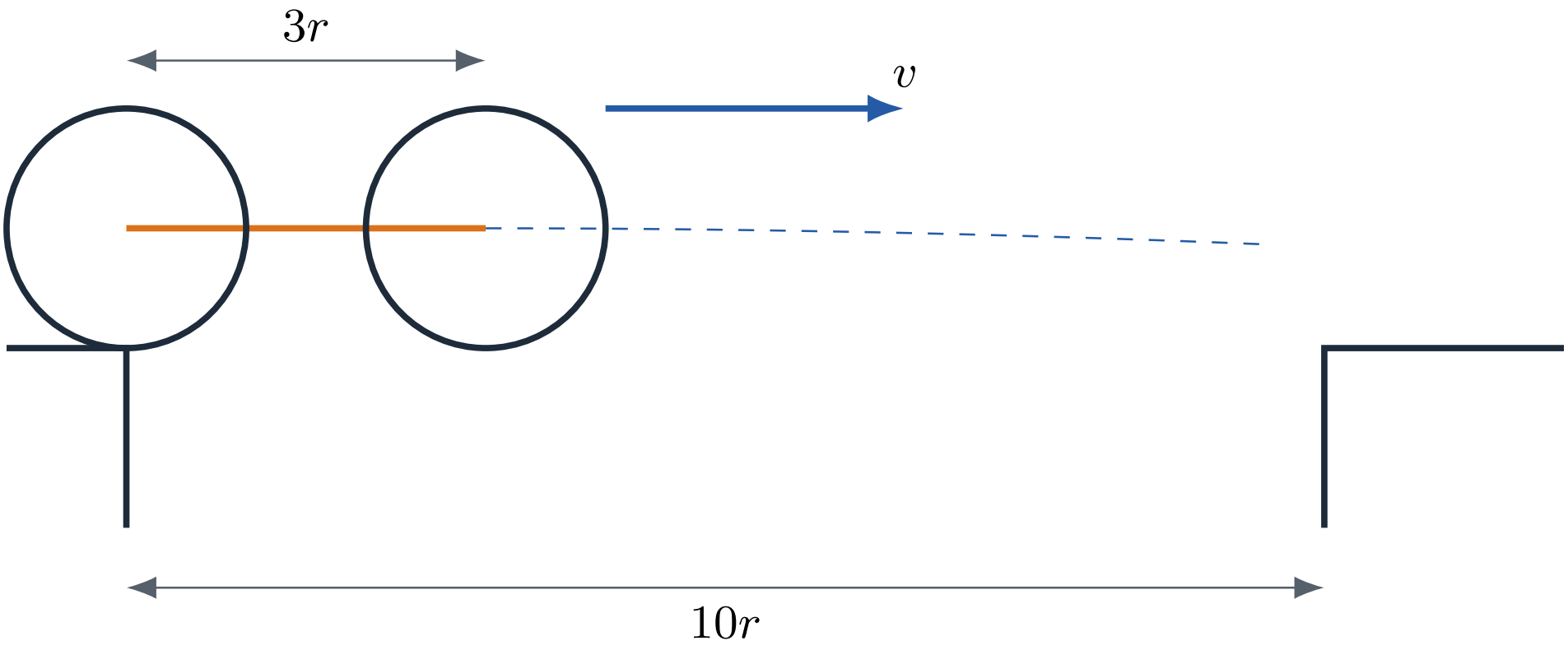


$$r = r_0 e^{\theta/\sqrt{3}}$$

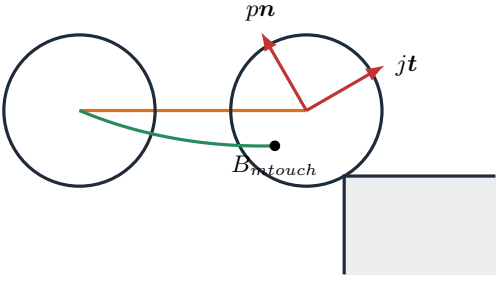
$$\mathrm{d}s = 2\mathrm{d}r$$

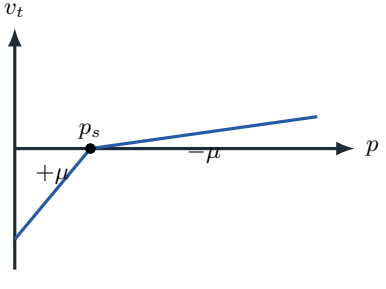
$$\boldsymbol{v}_{\text{abs}} = \boldsymbol{v}_{\text{rel}} + \omega r\boldsymbol{e}_\theta$$

$\omega < 0$ 时方位分量相减

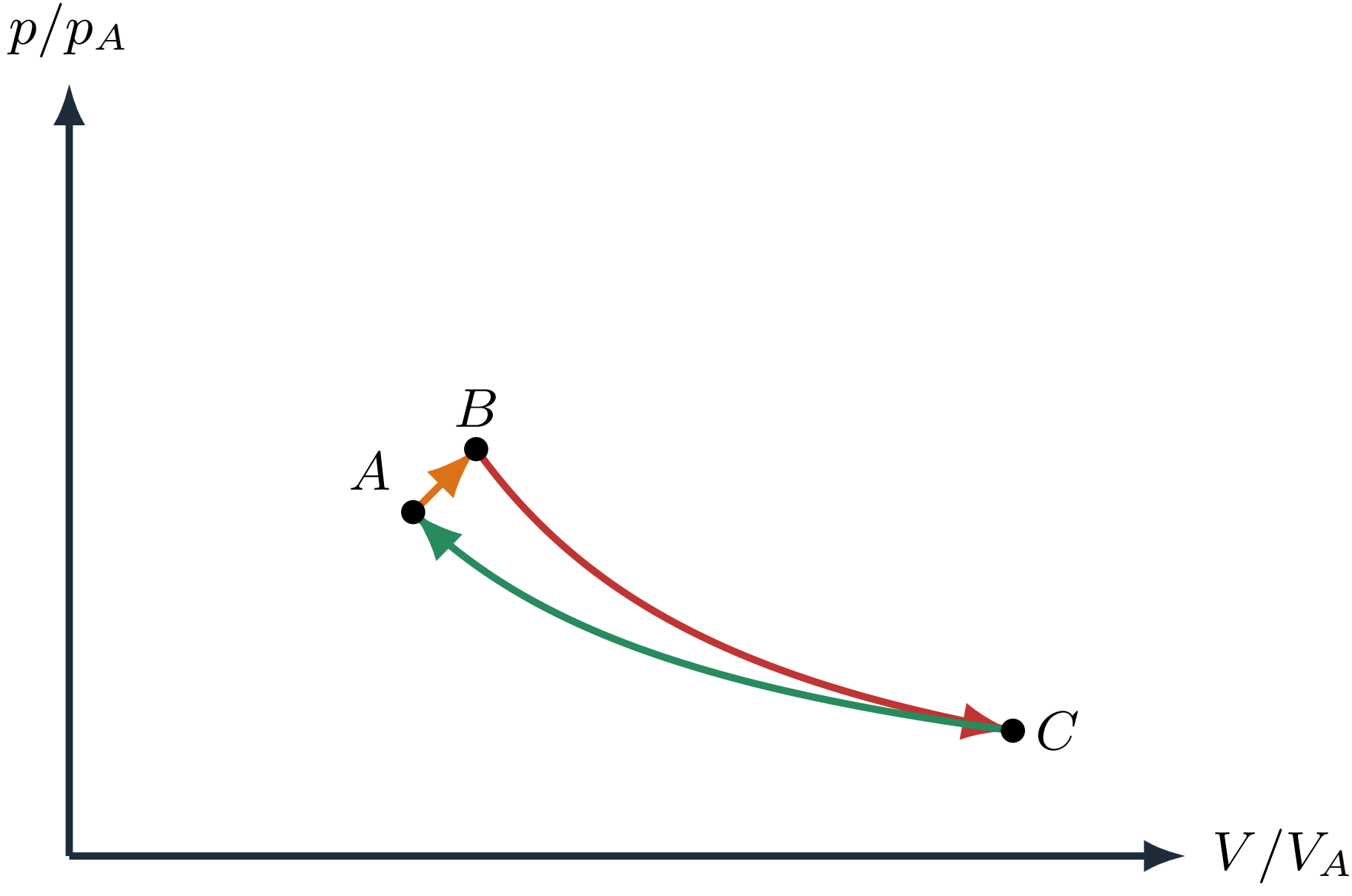


虚线为前轮轮心碰前平抛轨迹，半径均为 r

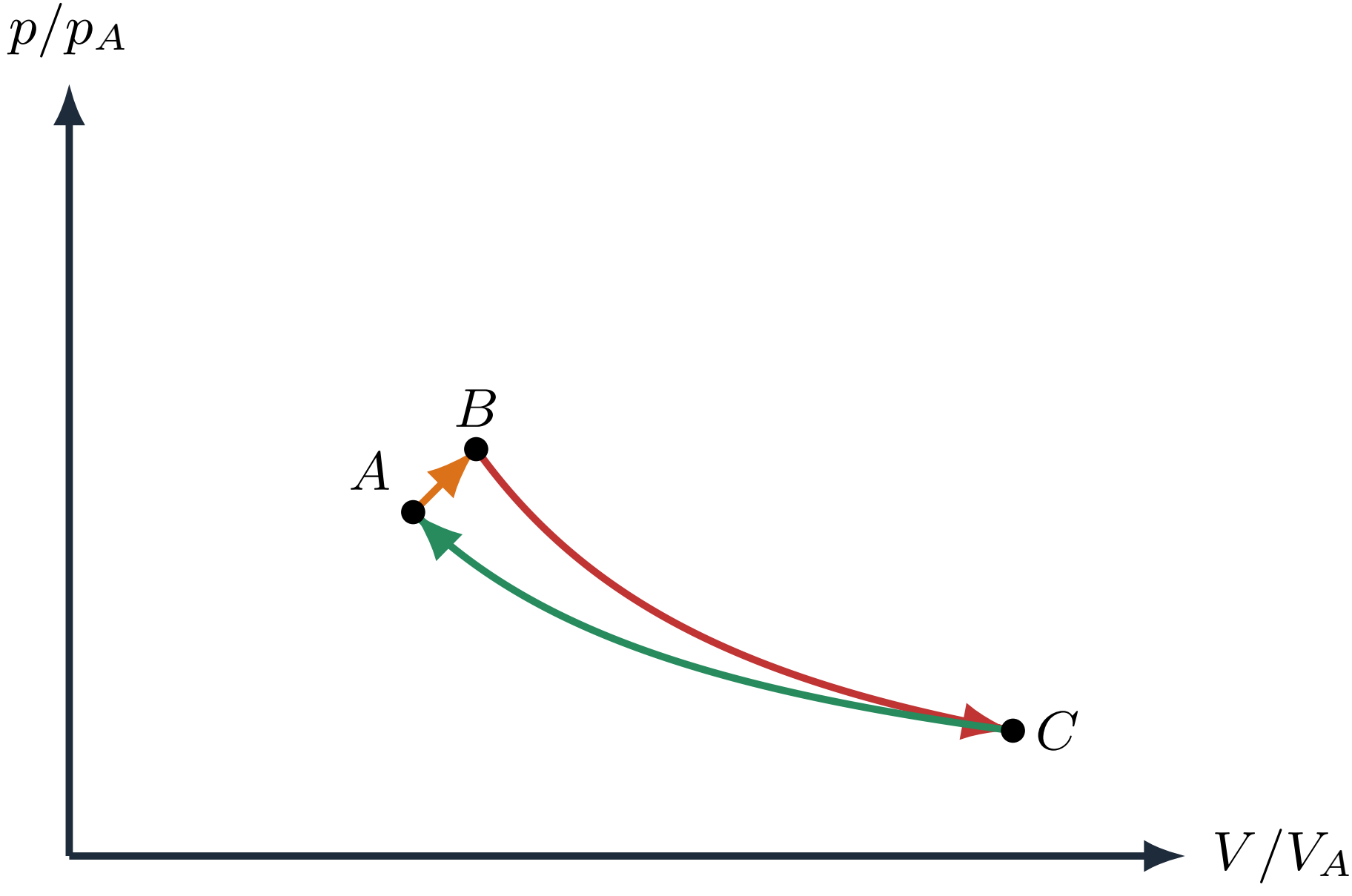




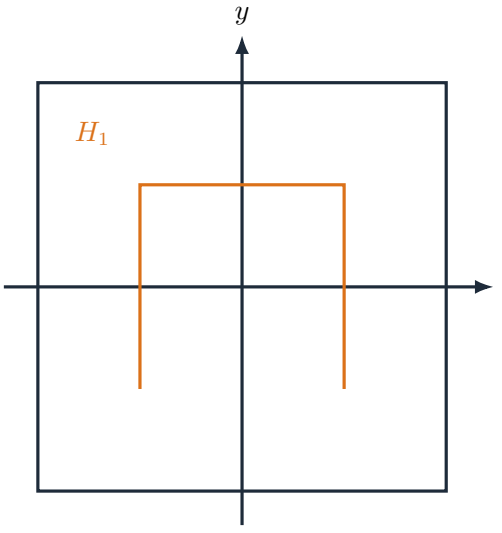
静摩擦不足，随后反向滑动

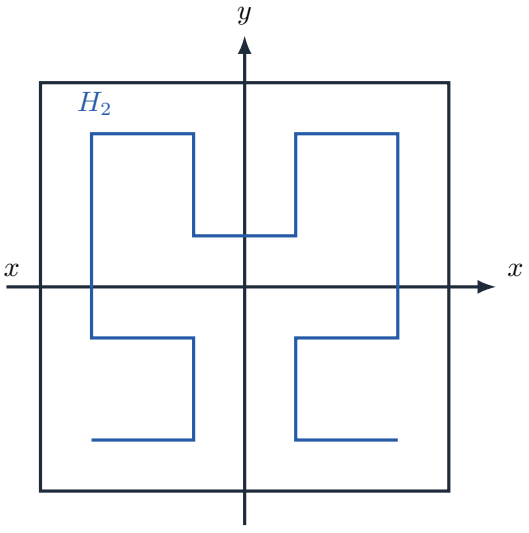


$T_2/T_1 = 1.4$ 示例；绿线严格满足 $pV = \text{const}$

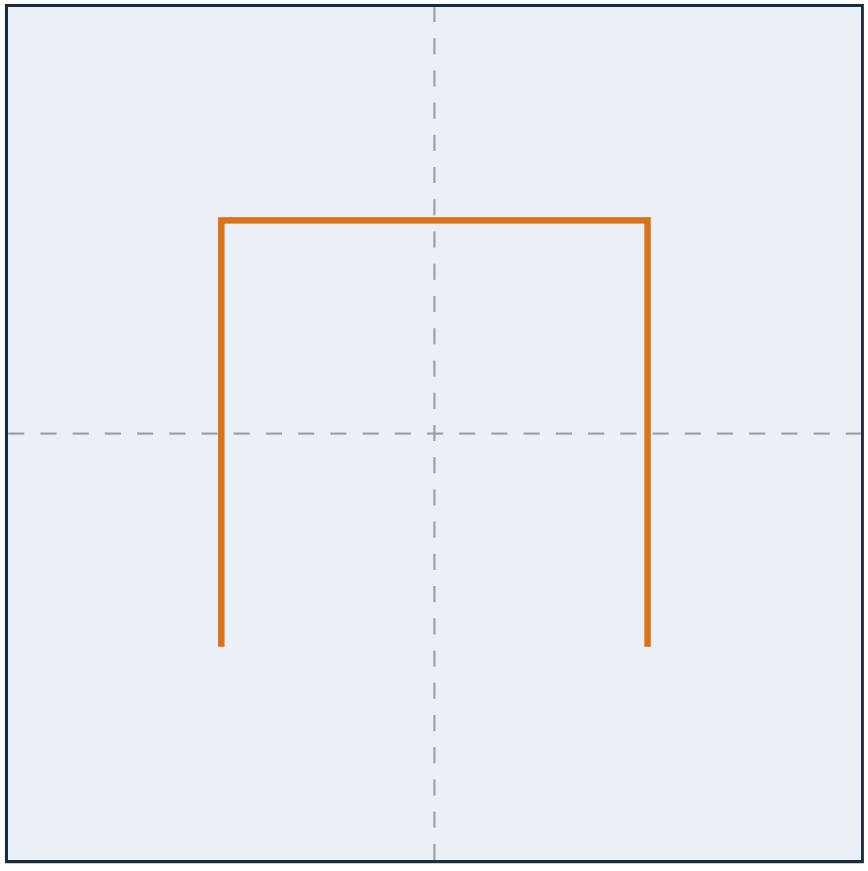


$T_2/T_1 = 1.4$ 示例；绿线严格满足 $pV = \text{const}$





外框边长均为 L，每阶曲线总质量均为 M



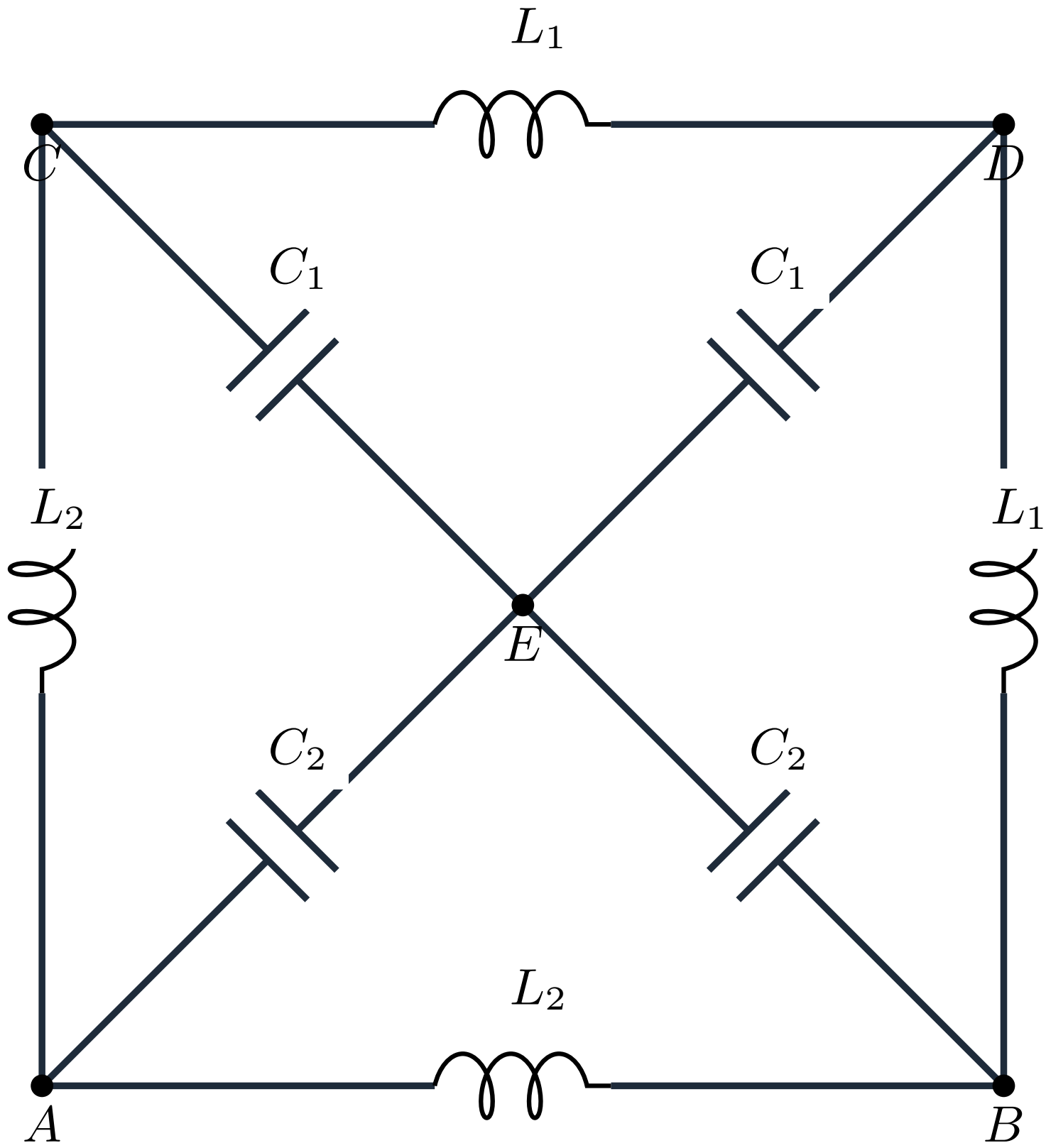
迭代并取极限

$$I_{1x} = \frac{5}{144}ML^2, \quad I_{1y} = \frac{7}{144}ML^2$$

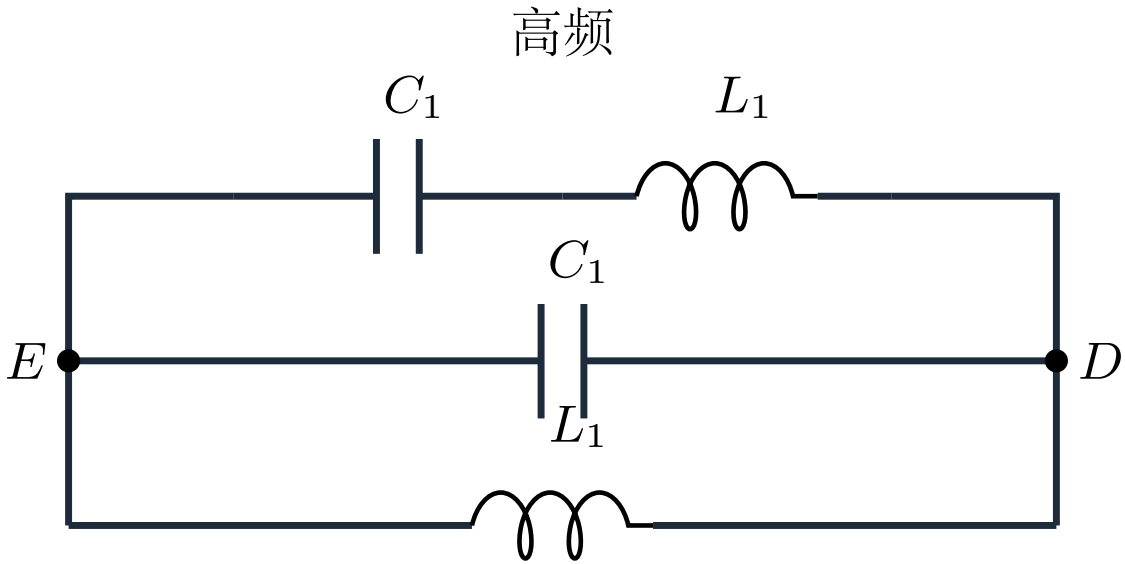
$$I_{2x} = \frac{197}{2880}ML^2, \quad I_{2y} = \frac{199}{2880}ML^2$$

归一化弧长测度趋于均匀面积测度

$$I_{\infty x} = I_{\infty y} = \frac{ML^2}{12}$$

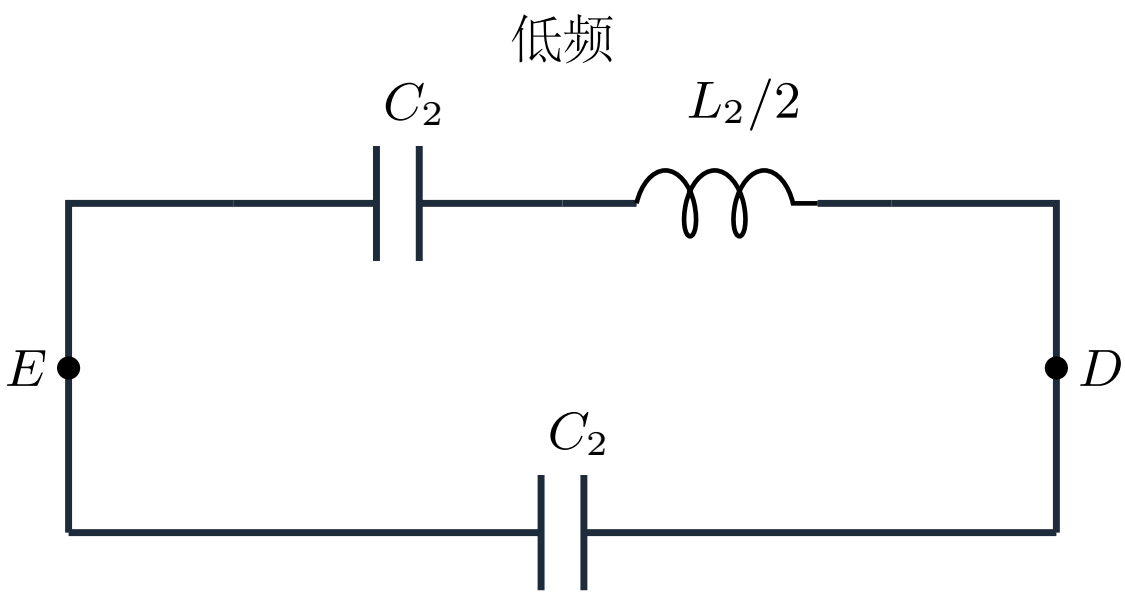


$$C_1 \ll C_2,\ L_1 \ll L_2$$

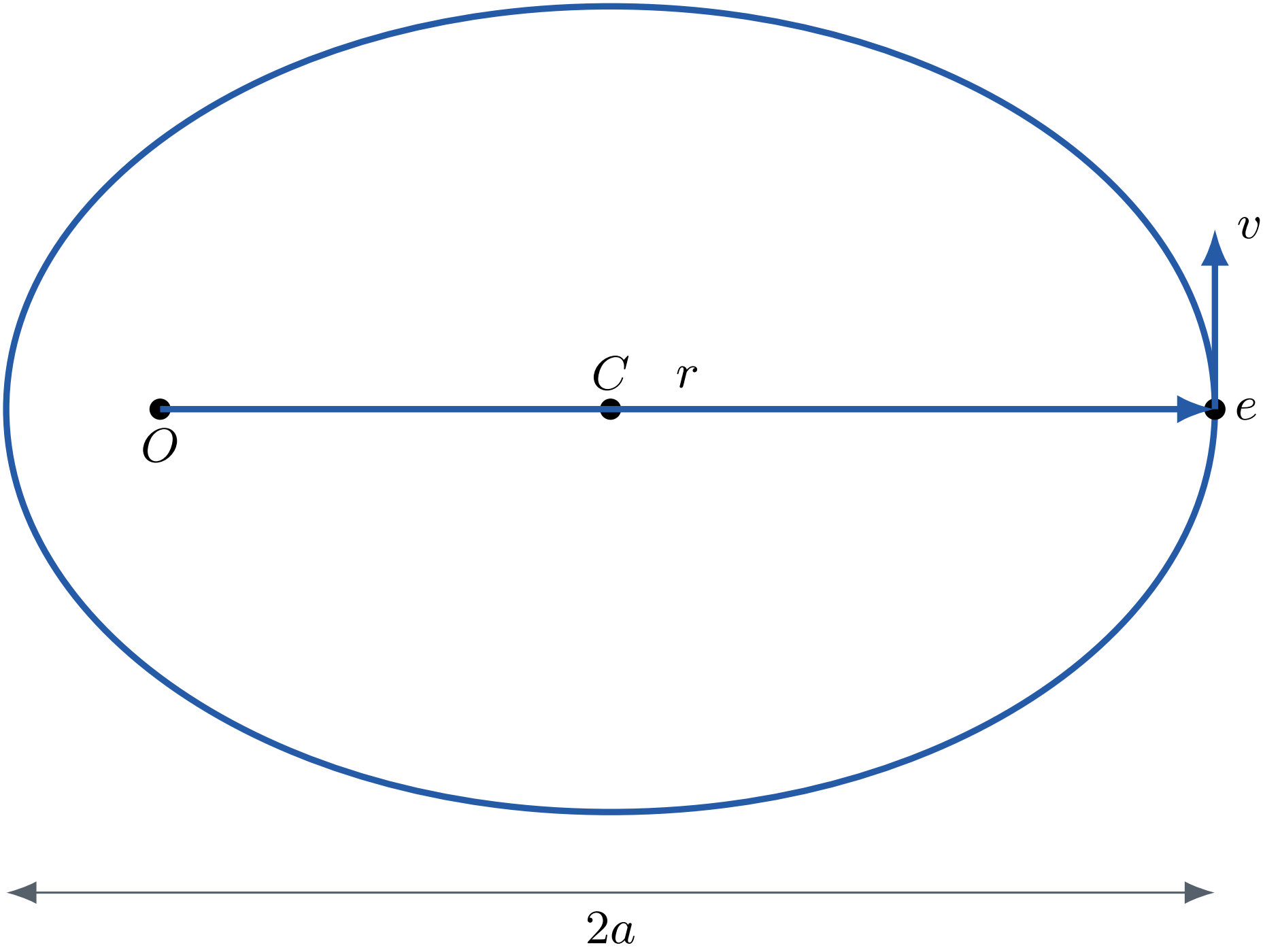


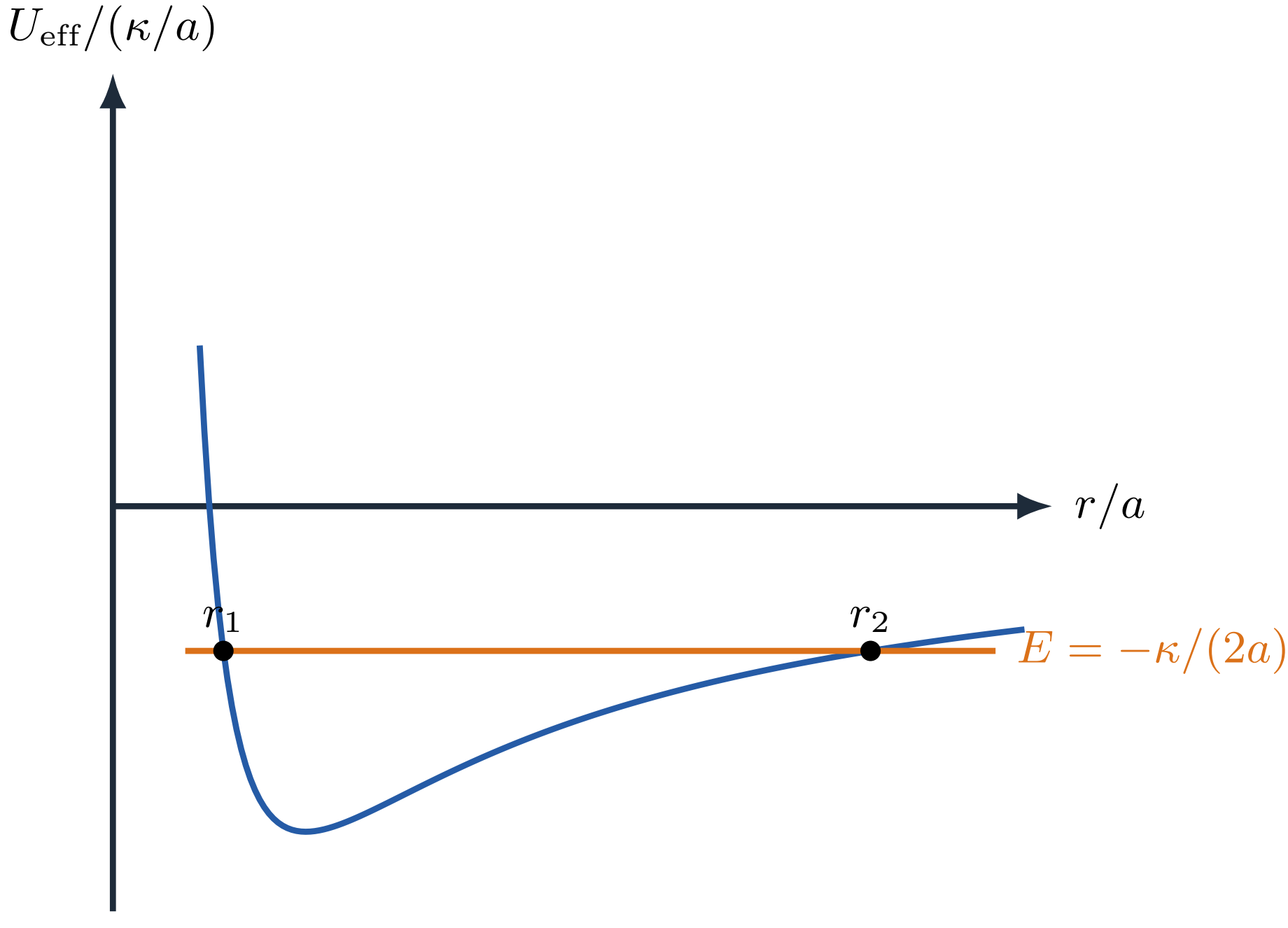
高频极点：$x^2 - 3x + 1 = 0$

低频极点：$y = 4$

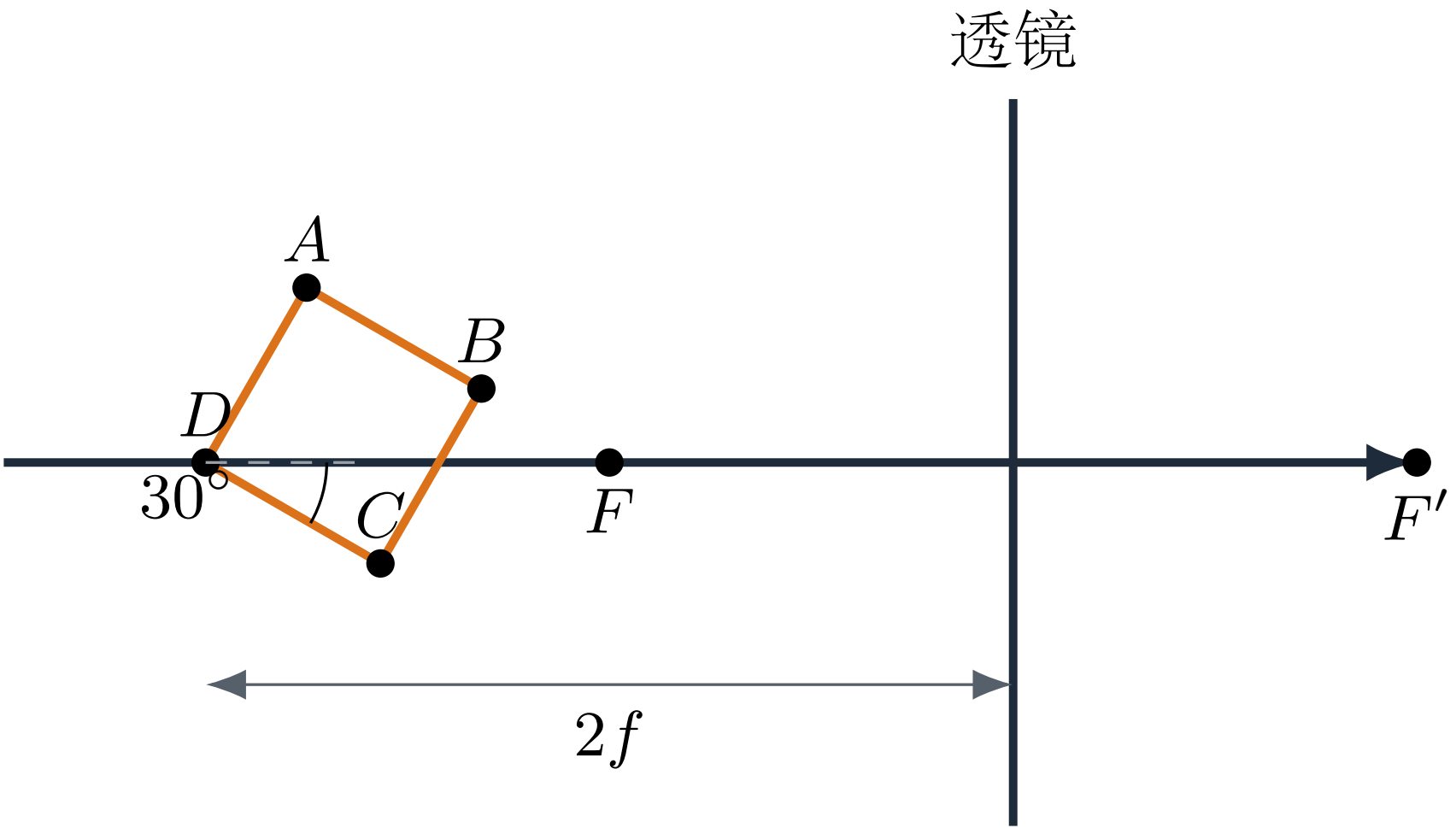


O 为焦点，C 为椭圆中心



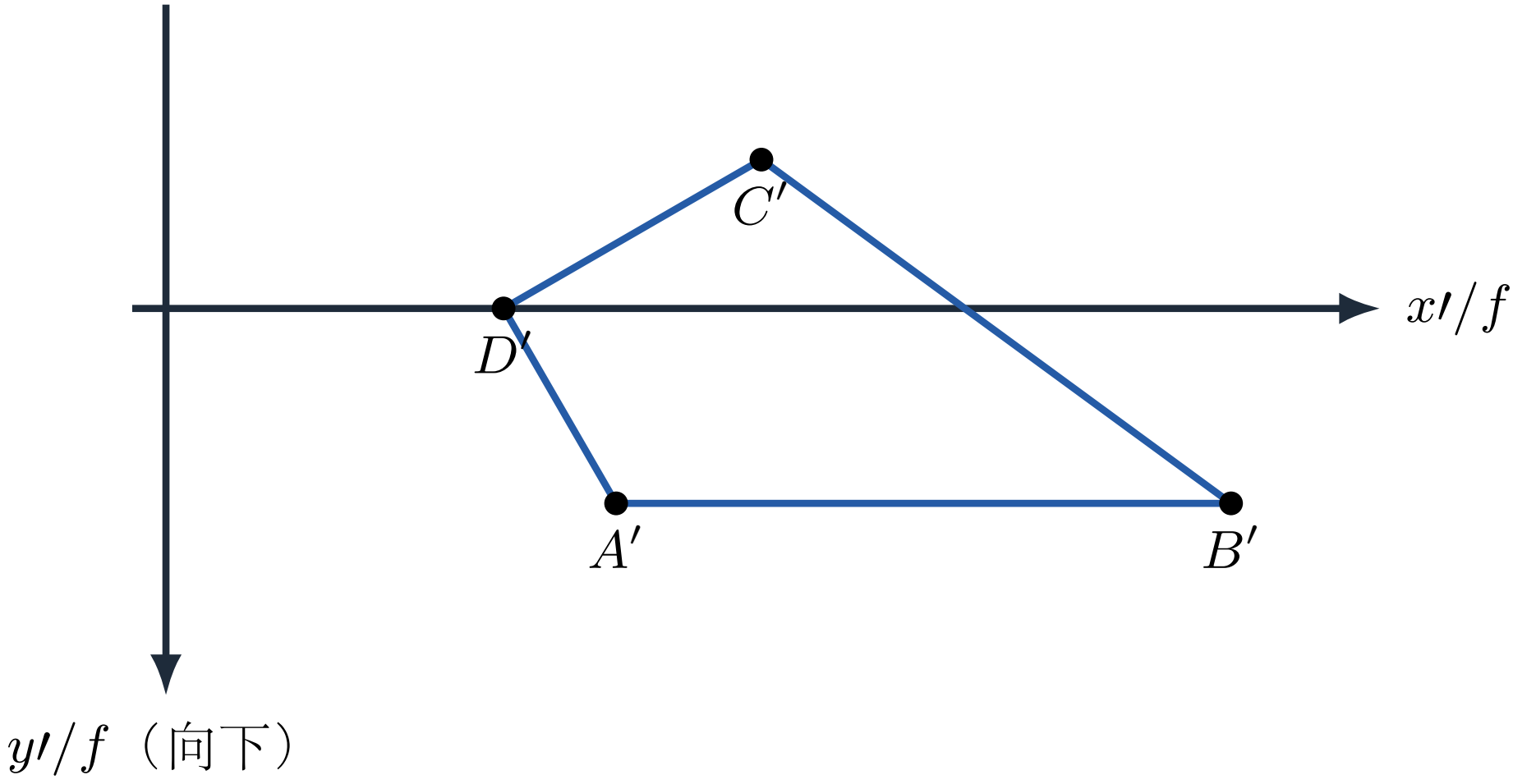


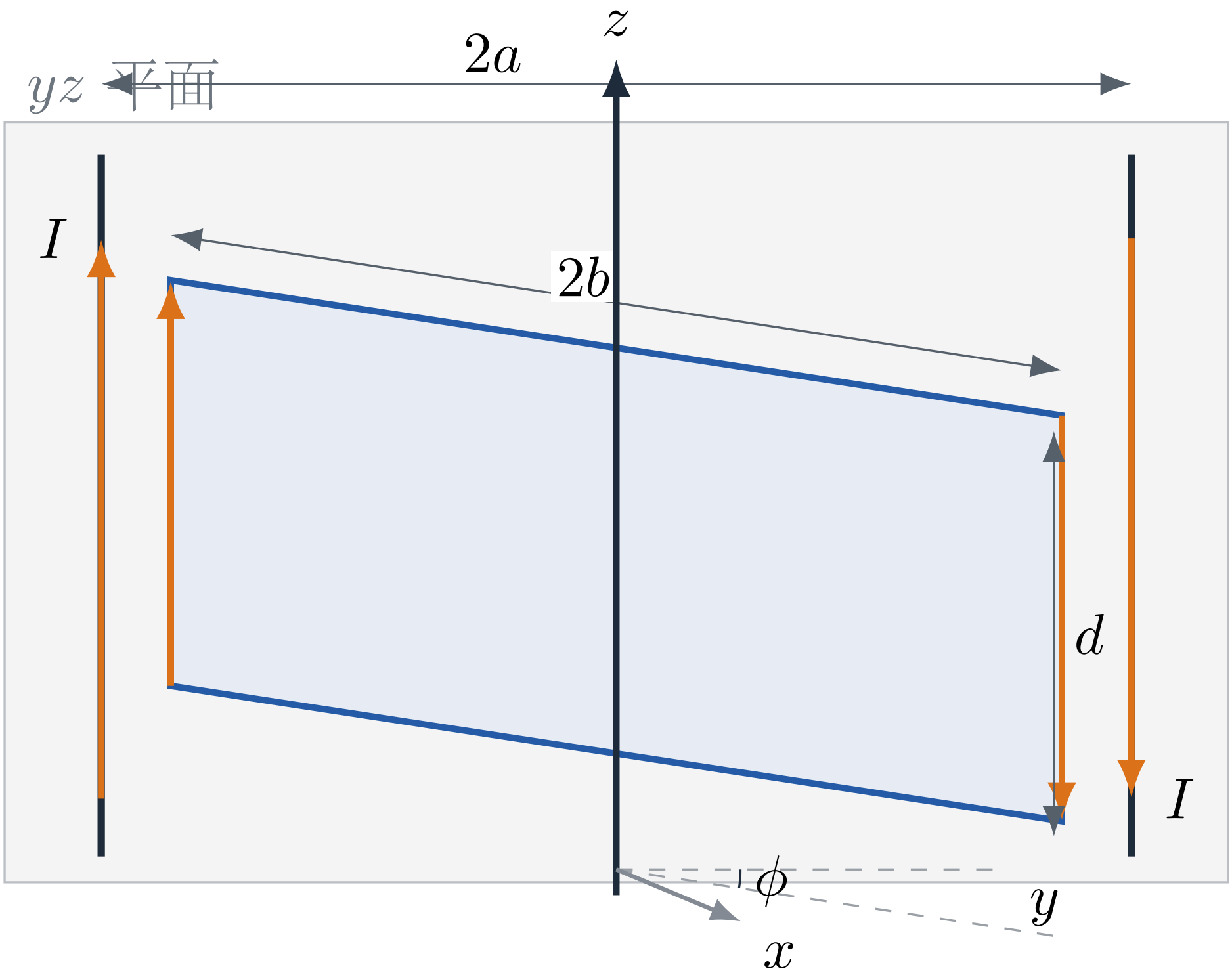
$$n = 3, l = 2 : \ \varepsilon = \sqrt{5}/3$$


透镜
A
B
D
30°
C
F
F′
2f

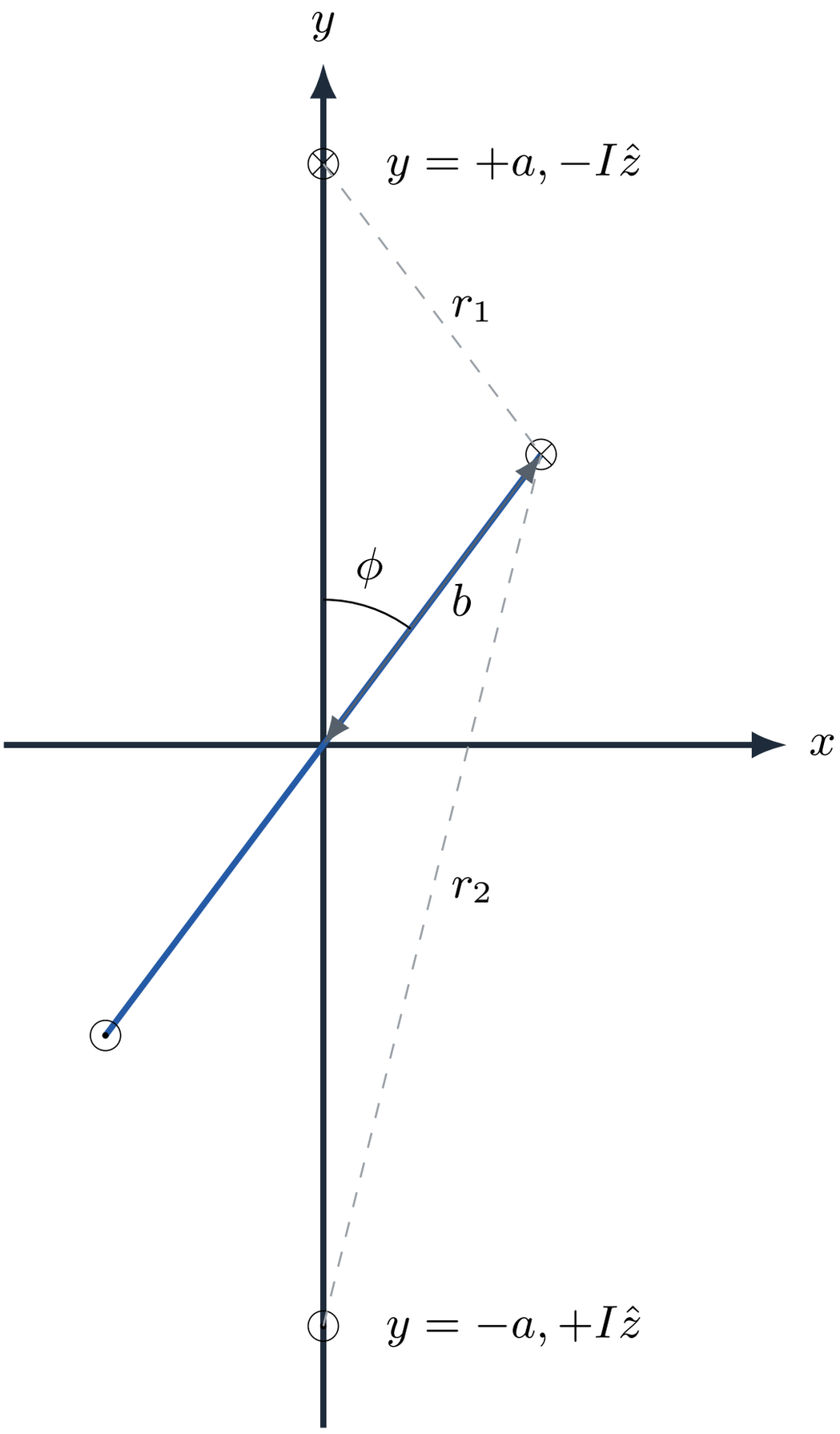
$$X\prime = 1/X,\ Y\prime = Y/X$$

$$S = (84 + 34\sqrt{3})f^2/117$$





矩形线框绕中心 z 轴转动



顶视图：$\boldsymbol{u} = (\sin\phi, \cos\phi)$

ϕ 为顺时针角

$Q_\phi = -M,\ \tau_z = +M$

$r_{1,2}^2 = a^2 + b^2 \mp 2ab\cos\phi$